

Lectures 13–14

Feeling the Wall Before You Reach It

Collision detection for haptic rendering — bounding volumes, and the fits they trade away

Robotics & VR — haptic simulation track

After Ericson, *Real-Time Collision Detection*, Ch. 4 · EECE527 Lectures 13–14

Bounding volumes · 90 minutes

THE LOOP WE ARE BUILDING

One Block of the Haptic Loop Decides the Rest

01

Read the hand

Encoders give the position and orientation of the stylus. Cheap, fixed cost.

02

Detect contact

Does the tool touch the scene? Cost grows with geometry, not with the device. **Today.**

03

Compute a force

Penetration depth and contact normal into a spring-damper response.

04

Drive the motors

The user's fingertips close the loop. They notice everything.



the user's fingertips are inside the loop – they close it

Blocks 1, 3 and 4 cost roughly the same every cycle. Block 2 is the only one whose cost depends on how complicated the world is — so it is the one that breaks the loop.

WHY • THE BUDGET

You Have One Millisecond. Every Millisecond.

A late frame is a dropped frame. A late force is a vibration in someone's hand — and the loop that produced it has a human closing it.

WHY • TOUCH IS NOT VISION

Sixteen Milliseconds Versus One

Graphic rendering

60 Hz → 16 ms

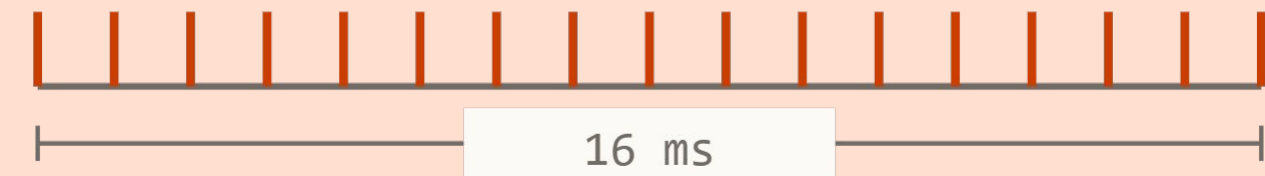


one force update per frame window

Vision integrates over tens of milliseconds. A missed frame is a stutter — annoying, recoverable, invisible if it happens rarely.

Haptic rendering

1 kHz → 1 ms



sixteen updates in the same window

The stiffness you can render stably rises with update rate. Too slow, and a rigid wall feels soft, then buzzy, then the device fights the user.

Same scene, same geometry, one sixteenth of the time — and the collision query is the term in that millisecond that grows with the scene.

THE PROBLEM • THE EXACT ANSWER IS UNAFFORDABLE

Every Triangle Against Every Triangle

	Virtual tool — a drill tip, a needle, a gripper finger	5×10^3 triangles
	Scene — the anatomy, the workcell, the cluttered bin	2×10^5 triangles
Exhaustive triangle pairs, per update		10^9 tests
At an optimistic 10 ns per triangle–triangle test		≈ 10 s
Budget		1 ms

brute force

1 ms budget

| $\approx 1/10,000$ of the bar above

Four orders of magnitude. Faster hardware does not close that gap — richer scenes widen it. The exact test cannot be what we run every cycle; it has to be what we run last, and almost never.

THE IDEA, BEFORE THE MATHEMATICS

Ask the Cheapest Question That Can Say No

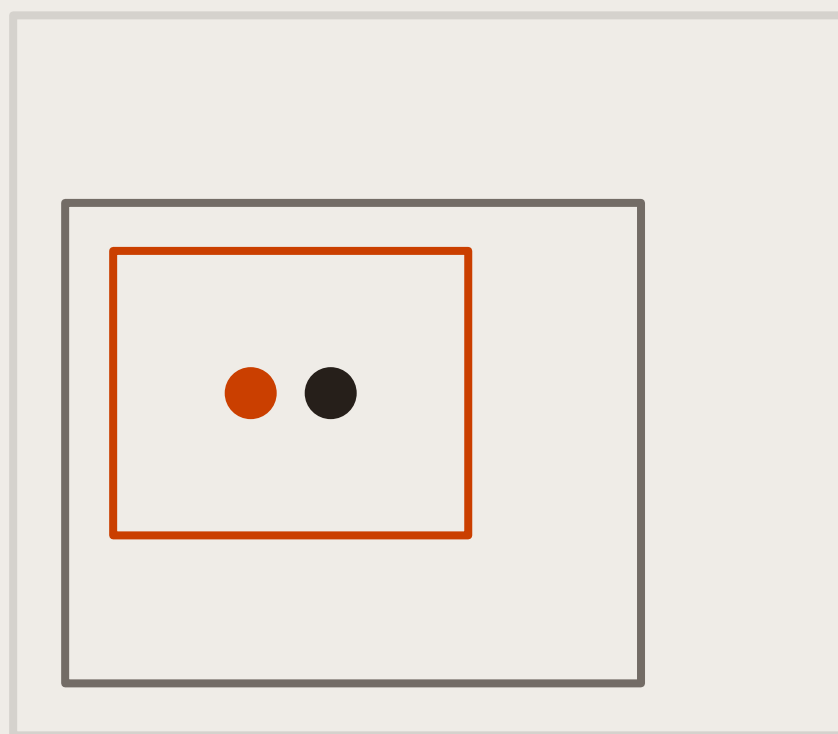
Are these two people in the same room?

Same city? 1 question

Same district? 1 question

Same building? Same floor? 2 questions

Walk the corridor and look expensive



city $\supset$ building $\supset$ room one
containment test per level

The asymmetry is the whole trick

A **no** at any level is final, and it cost one comparison.

A **yes** proves nothing — it only buys the right to ask a more expensive question.

In a haptic loop, almost every query is a no. Optimize for the answer you actually get.

LECTURE 13 • BOUNDING VOLUMES

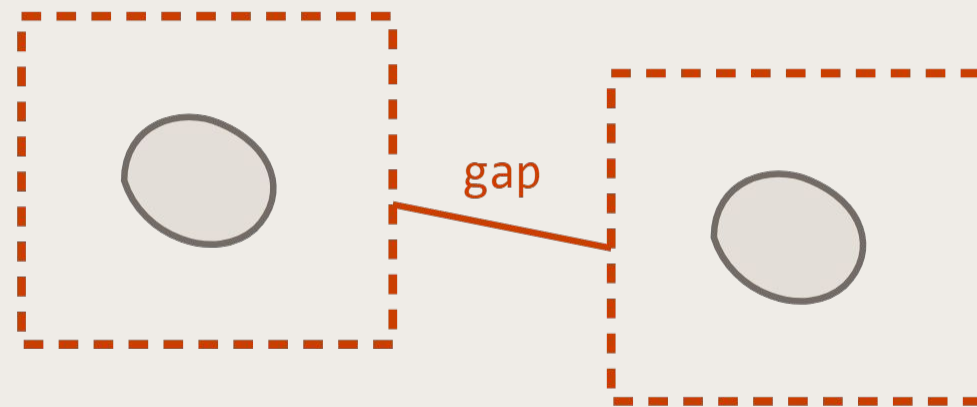
Wrap It in Something Stupid and Fast

A single simple volume encapsulating one or more objects of more complex nature — bought for one reason: fast overlap rejection.

WHAT THE TEST ACTUALLY PROVES

No Means No. Yes Means Maybe.

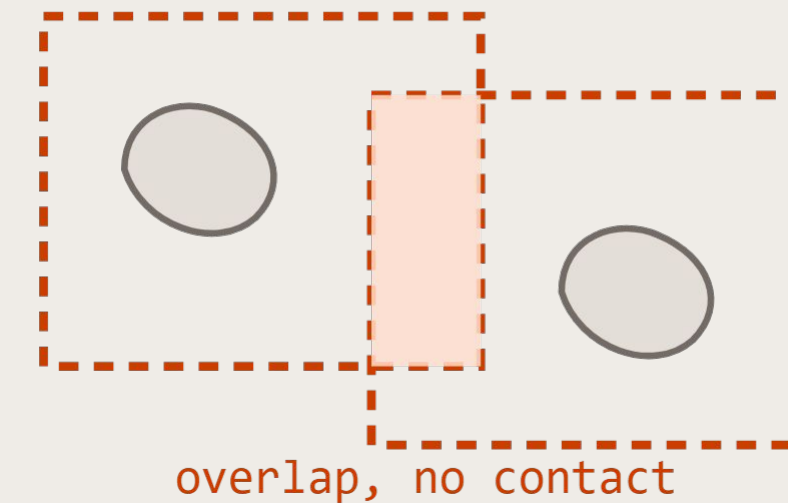
BVs disjoint



The objects inside are **certainly** not in contact — the only exact conclusion a bounding volume gives you. Stop here.

no false negatives

BVs overlap



You have learned nothing about the objects. Descend a level, or run the exact test.

false positives cost time

Tightness never buys correctness. It buys the *frequency* of the expensive path — which, at 1 kHz, is the same thing as buying stability.

LECTURE 13 • DESIRABLE BV CHARACTERISTICS

Five Requirements That Fight Each Other

01

Inexpensive intersection tests

Charged on every query, at every level of the hierarchy.

02

Tight fitting

Loose wrappers say “maybe” too often and push work downward.

03

Inexpensive to compute

Sometimes offline — but not for deformable or freshly loaded geometry.

04

Easy to rotate and transform

The item haptics cares about most: our tool moves every cycle.

05

Use little memory

A hierarchy stores one volume per node — thousands of them.

No volume wins all five. The convex hull is tightest and fails the rest; the sphere is cheapest and fits a long thin tool badly. Everything that follows is a choice of which requirement to sacrifice.

LECTURE 13 • AXIS-ALIGNED BOUNDING BOXES

Six Numbers, and a Promise Not to Rotate

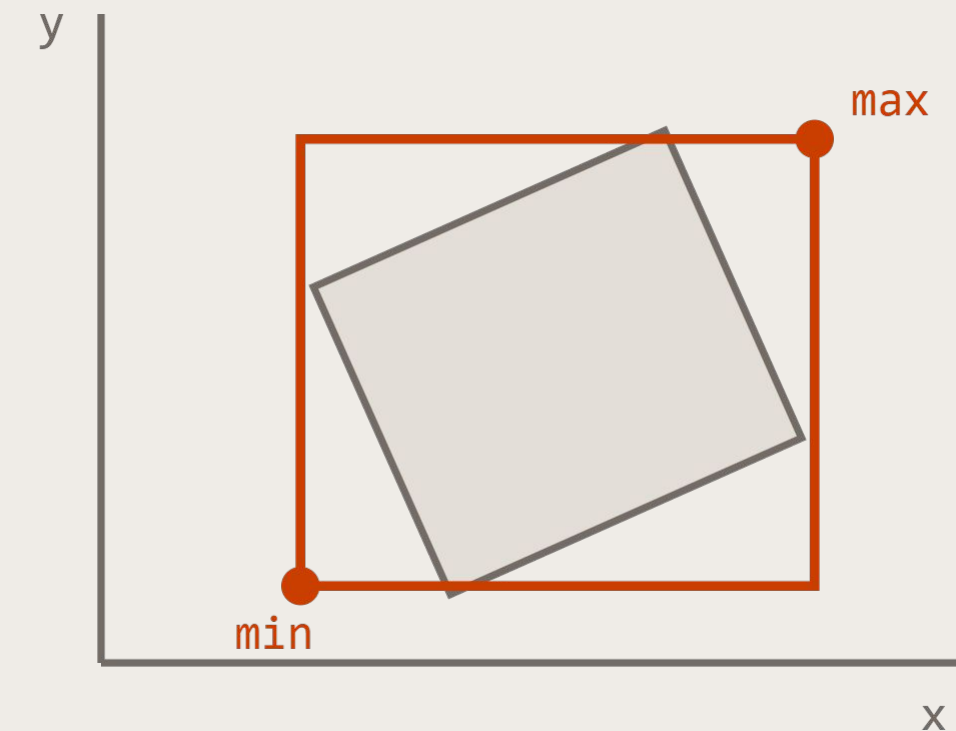
A rectangular six-sided box whose faces are oriented so that its face normals are always parallel with the axes of the given coordinate system.

Three equivalent representations

min-max • min-widths • centre-halfwidths

Six floats either way — the cheapest box you can store.

Leads to very fast overlap checks, and is one of the most common bounding volumes in use.



two points, six numbers — the box never tilts

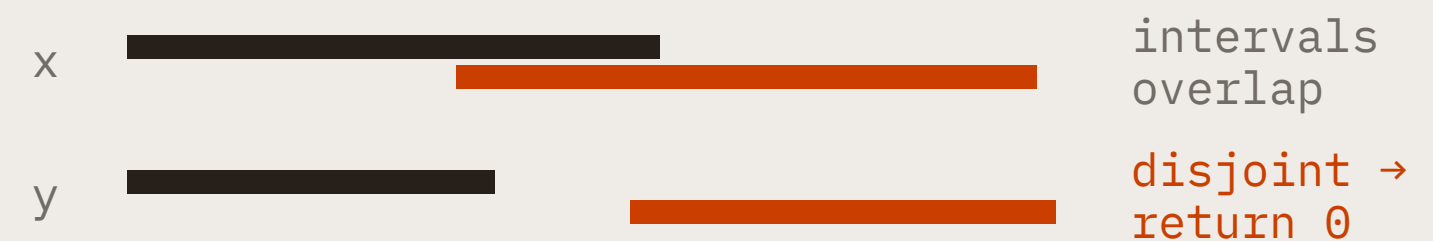
LECTURE 13 • AABB-AABB INTERSECTION

Three Intervals, and an Early Exit

```
int TestAABBAABB(AABB a, AABB b)
{
    if (a.max[0] < b.min[0] ||
        a.min[0] > b.max[0]) return 0;
    if (a.max[1] < b.min[1] ||
        a.min[1] > b.max[1]) return 0;
    if (a.max[2] < b.min[2] ||
        a.min[2] > b.max[2]) return 0;
    return 1;
}
```

Two AABBs overlap only if they overlap on all three axes

Each dimension's extent is just an interval on its axis. Separation on one axis is separation, full stop.



Why it suits a 1 kHz loop

Comparisons only — no multiplies, no square roots — and it returns early on the answer you get 99% of the time.

THE PROBLEM WITH AXIS ALIGNMENT

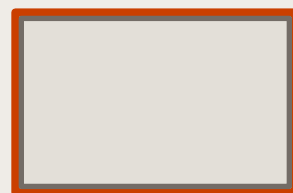
Nobody Holds a Stylus Still

Freely oriented volumes

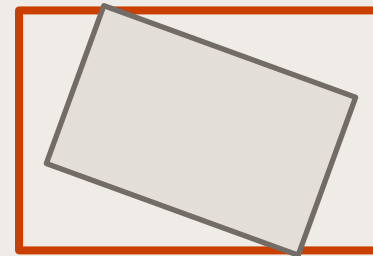
Spheres and convex hulls naturally transform into any coordinate system — they are not restricted to specific orientations. A rotation costs them nothing.

Aligned volumes

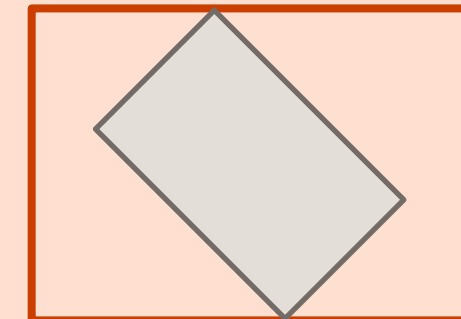
AABBs are restricted in the orientations they can assume. Object rotation leaves them unaligned, so they must be **realigned** — recomputed — as the object moves.



0° — tight



20° — slack appears



45° — rebuilt, and looser

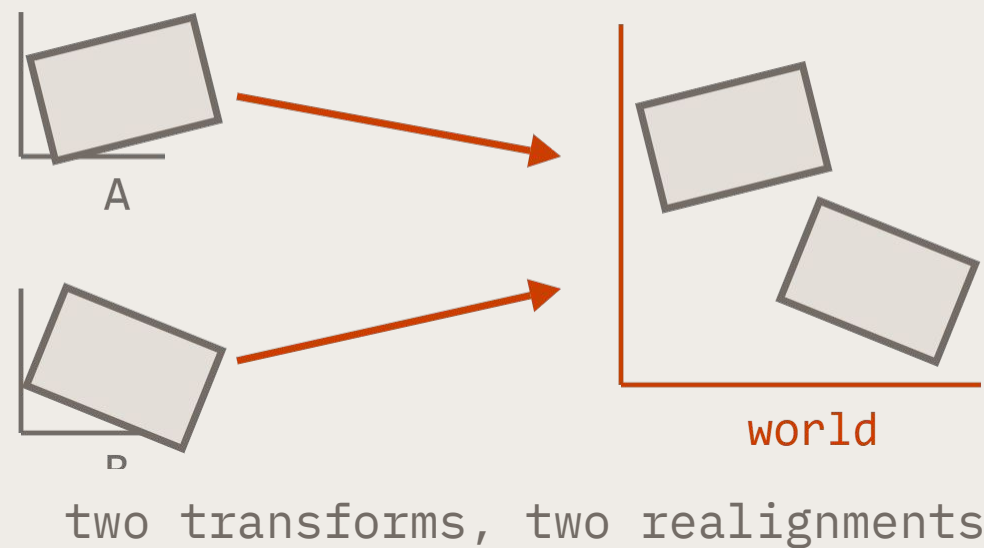
In graphics you pay realignment occasionally. In our loop the tool rotates in someone's fingers a thousand times a second — and the bill arrives whether or not a contact is anywhere nearby.

LECTURE 13 • COMPUTING AND UPDATING AABBS

Move One Volume, Not Both

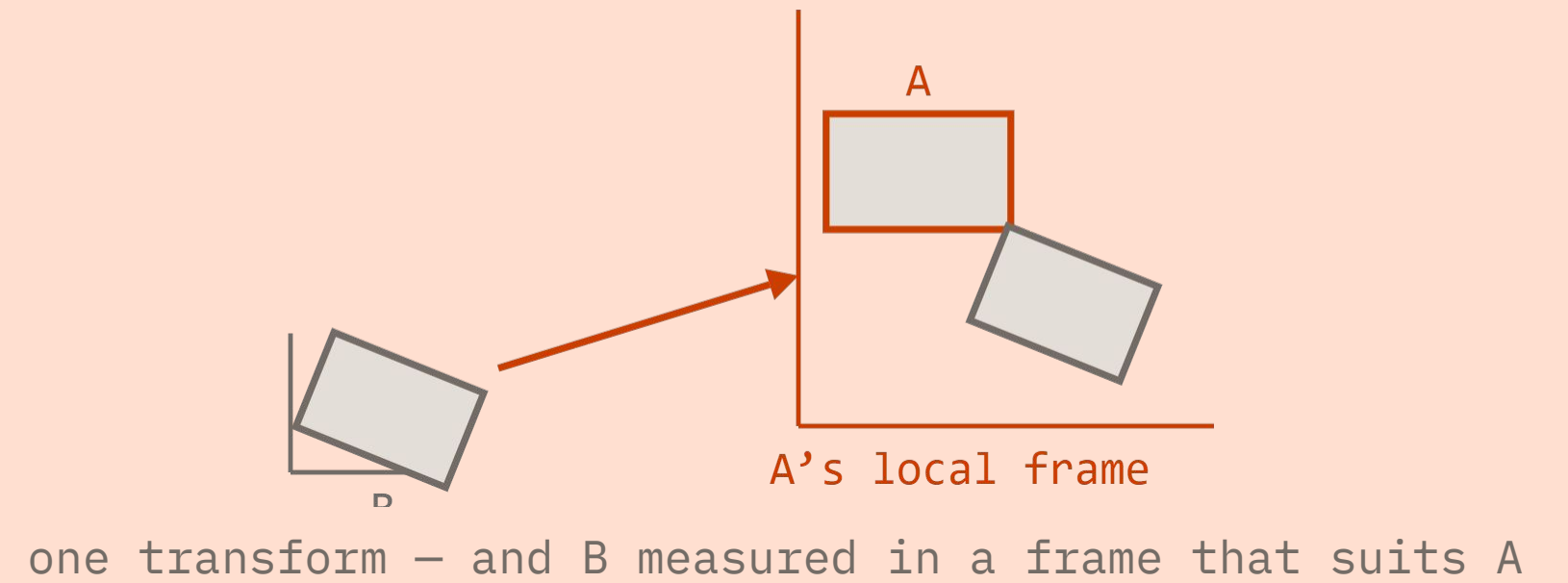
Both into world space

The obvious reading of “put them in a common frame.” Two transforms, two realignments, two chances to inflate the volume.



One into the other's local space

Preferred. The required computation is half of the former, and it often results in a tighter bounding volume.



Same habit returns in Lecture 14: expressing B in the coordinate frame of A is what makes the OBB separating-axis test affordable.

LECTURE 13 • FOUR WAYS TO REALIGN

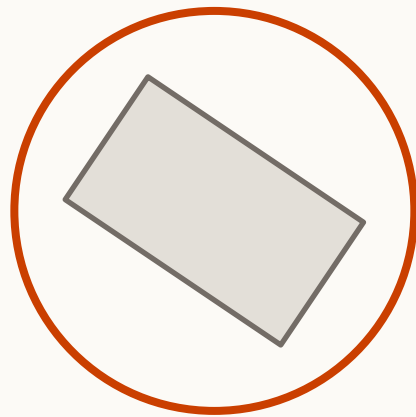
Rebuild It, or Refuse to Rebuild It

Method	How it works	Cost	Fit
 AABB of the bounding sphere	One fixed-size box large enough to contain the object at any orientation. Completely circumvents the need to reshape the AABB.	none	loosest
 Rebuild from the point set	Find six extreme points, two per axis. Concave object → convex hull first, its k vertices stored in the first k elements. An $O(\log n)$ variant exists using a precomputed search structure.	$O(n)$	tight
 Hill climbing on neighbours	Keep six vertex pointers at the extremal vertices; each update, compare them against neighbour vertices to see if they are still extremal. Needs convex geometry — so run it on a precalculated convex hull.	expected $O(1)$	tight
 Wrap the rotated AABB	Most common approach. Approximate, not tight — and it must be computed from a rotation of the <i>original local-space</i> AABB, or the error compounds every update.	$O(1)$	approximate

LECTURE 13 • BOUNDING SPHERES

The Volume That Does Not Care How You Turn It

Rotationally invariant



the same sphere at every orientation

Trivial to transform — the sphere simply has to be translated to its new position.
No realignment, ever.

Most memory-efficient

A centre and a radius: four floats per node in the hierarchy, against six for an AABB.

Inexpensive test

```
d = b.c - a.c;  
r = a.r + b.r;  
return dot(d,d) <= r*r;
```

Square both sides and the square root disappears.

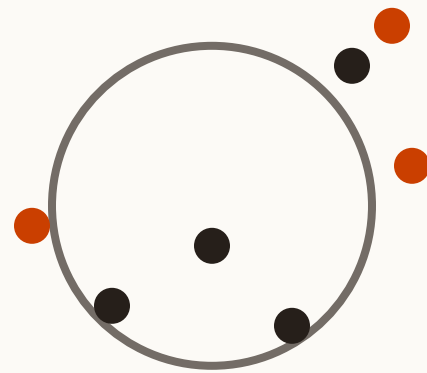
The catch, and the subject of the next four slides: computing an *optimal* bounding sphere is not as easy as computing an optimal AABB.

LECTURE 13 • COMPUTING A BOUNDING SPHERE

Guess Badly, Then Grow It Until It Fits

pass 1

Find a good initial almost-bounding sphere

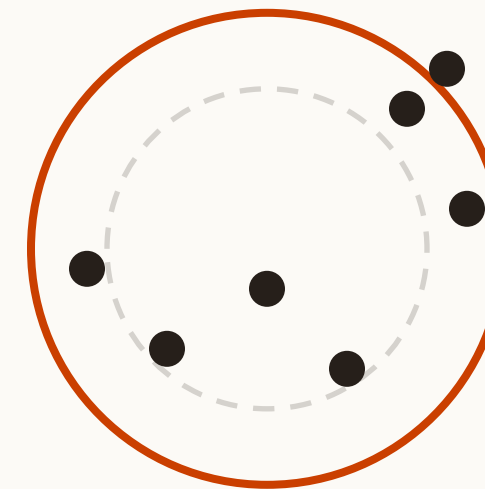


three points still outside

Cheap and approximate. It is not required to contain every point — only to start close.

pass 2

Improve the sphere until it bounds all points



grown just enough — dashed = pass 1

Any point outside grows the sphere by the minimum needed. Results in a tight fit.

Everything now hinges on pass 1 — a better initial sphere yields a better final sphere. That is what the next two methods buy.

LECTURE 13 • BOUNDING SPHERE FROM MAXIMUM SPREAD

Let the Point Cloud Tell You Its Long Axis

1 • Covariance

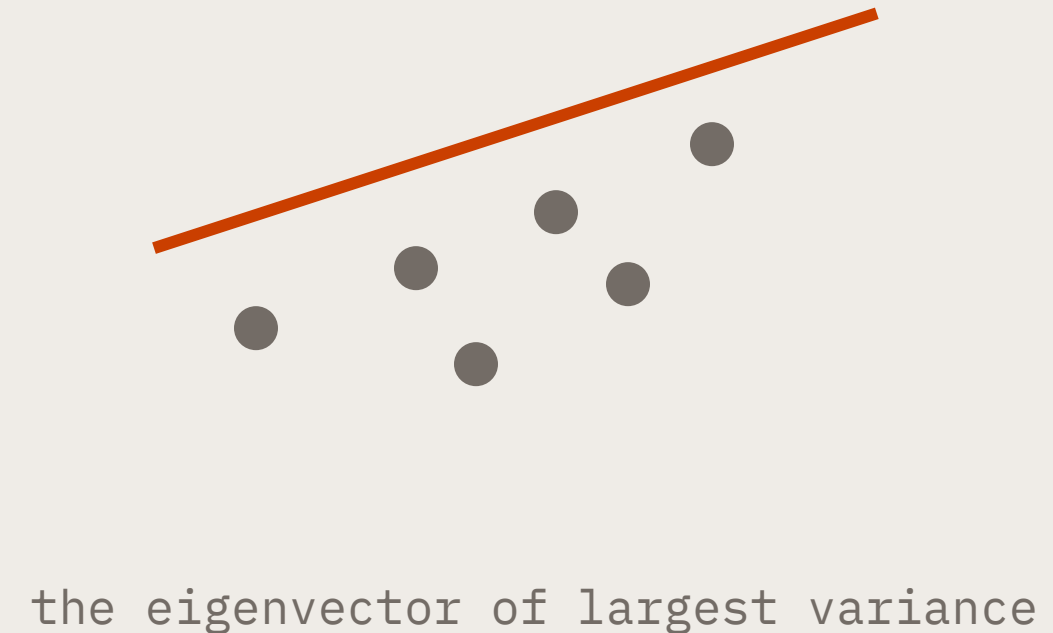
Identifies how the variables co-vary. A 3×3 symmetric matrix built from the point set.

2 • Eigen-decomposition

Largest-magnitude eigenvalue $\rightarrow$ the axis of largest variance.
Smallest $\rightarrow$ the axis of least variance.

3 • Sphere from two extremes

Project onto that axis; the two most distant projected points give the centre and radius of the starting sphere.



Learn this properly now: Lecture 14 uses the same decomposition to decide which way a box should point.

LECTURE 13 • ITERATIVE REFINEMENT

Feed the Answer Back In as the Guess

A better initial sphere gives a better final sphere — so run the algorithm on its own output, iteratively.

Shrink each result by a small amount so it becomes an underestimate for the next call — otherwise there is nothing left to refine.

Randomise the point order between iterations instead of visiting them the same way every time.

What it buys



Usually a much better sphere than the maximum-spread method, for a few extra passes over the input.

Apply the same iteration to the maximum-spread method and the two become comparable — the iteration was doing the work, not the statistics.

LECTURE 13 • THE MINIMUM BOUNDING SPHERE

The Exact Answer, and Why Nobody Brute-Forces It

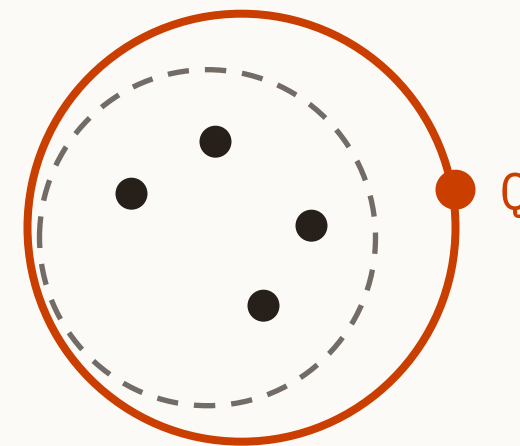
Brute force

A sphere is uniquely defined by four non-coplanar points. Try every combination of four — then three, then two — keep each sphere that contains all other points, and take the smallest.

$O(n^5)$

Welzl (1991) — expected linear time

Given a minimum bounding sphere S for a point set P , adding a point Q forces a recomputation **only if Q lies outside S** — and then Q must lie on the boundary of the new minimum sphere for $P \cup \{Q\}$.



dashed = S for P

expected $O(n)$

Homework (Lecture 13): take an example that illustrates how `WelzlSphere()` works, explaining its recursive behaviour step by step. Due one week later.

LECTURE 14 • BOUNDING VOLUMES (2)

Buying a Tighter Fit, and Paying For It

A box that can rotate. A volume shaped like the tool. A polytope that approaches the convex hull. Each one spends test cost, memory, or update cost — watch which.

LECTURE 14 • ORIENTED BOUNDING BOXES

Let the Box Turn With the Tool

Five representations — only one of them is worth using

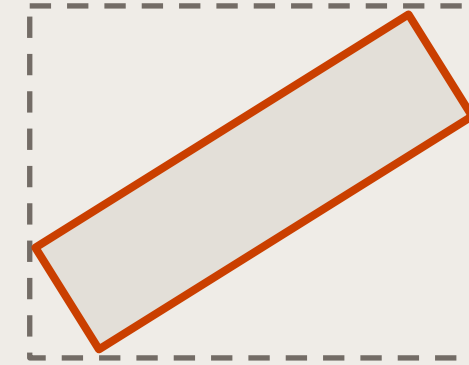
eight vertices

six planes

three slabs — pairs of parallel planes

a corner vertex plus three orthogonal edge vectors

a centre point + an orientation matrix + three half-edge lengths — **preferred**, because it makes the OBB–OBB test much cheaper



dashed AABB vs. the OBB that fits — same object

The memory bill

15 floats = 60 bytes

Against 6 floats for an AABB and 4 for a sphere — an expensive volume, multiplied by every node of a hierarchy.

Why not a quaternion?

Euler angles or quaternions store the orientation in three or four components instead of nine — but the separating axis test needs a matrix, and converting back is a very expensive operation.

LECTURE 14 • OBB-OBB INTERSECTION

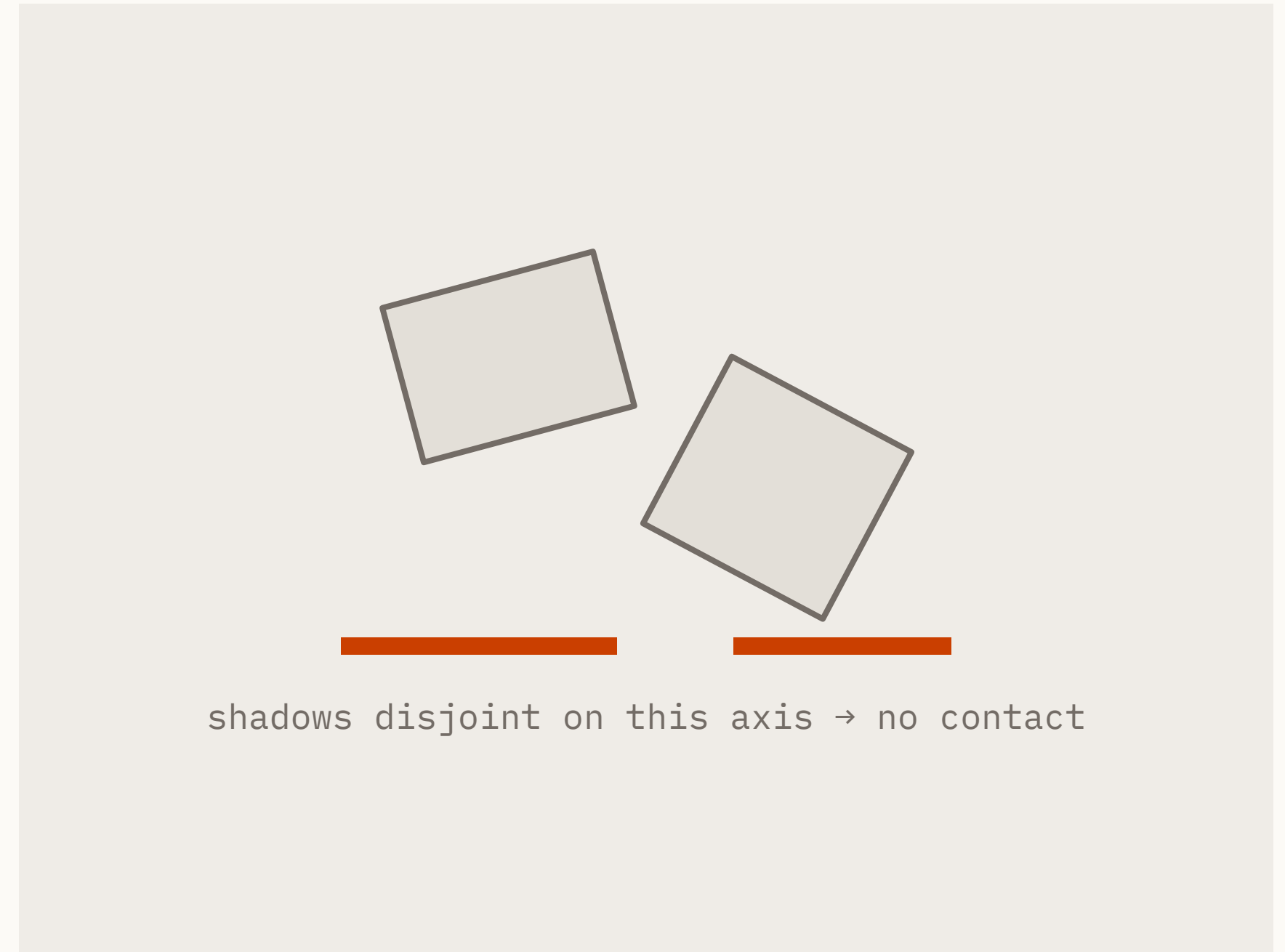
Find One Direction Where the Shadows Miss

Two objects A and B are **separated** if there exists some axis L on which their projections do not overlap.

Separated along L when the projected centre distance exceeds the sum of the projected radii:

$$|T \cdot L| > r_A(L) + r_B(L)$$

One witness axis ends the query. No witness among the candidates means they intersect.



The AABB test was this same test on three fixed axes. That is the entire reason it was cheap.

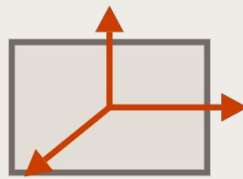
LECTURE 14 • THE FIFTEEN CANDIDATE AXES

Fifteen Axes Is the Worst Case, Not the Cost

3

Coordinate axes of A

Face normals of the first box.

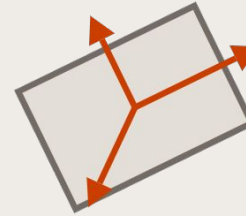


A's three face normals

3

Coordinate axes of B

Face normals of the second box.

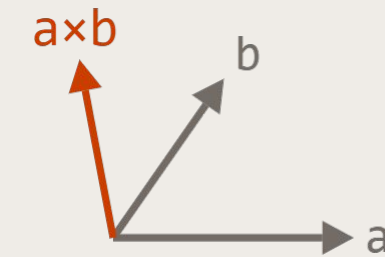


B's three, pointing anywhere

9

Cross-product axes

Perpendicular to one axis of A and one of B.



nine such products

Express B in A's frame. Fewer operations per axis — the entries of the rotation matrix R do the work.

Test in the prescribed order. One separating axis ends the query — ordering decides whether the common case costs two tests or fifteen.

Note the failure mode for us: the deeper the contact, the more axes survive — the test is slowest exactly when the user is pressing hardest.

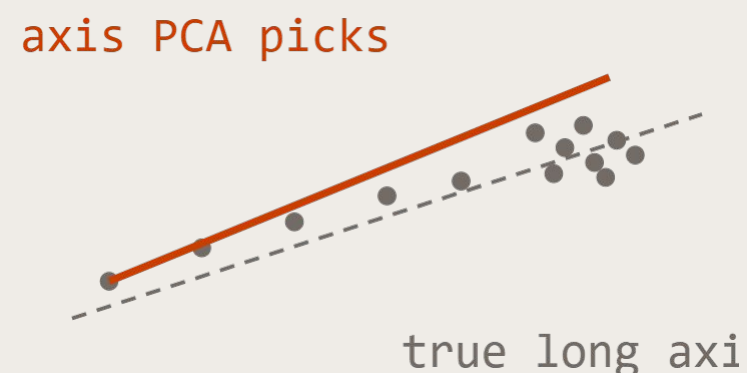
LECTURE 14 • COMPUTING A TIGHT OBB

There Is No Exact, Practical Algorithm

The volume difference between a poorly aligned and a well-aligned OBB can be large — get the orientation wrong and you lose a lot, not a little.

PCA, and its trap

Covariance finds the longest axes — but it depends heavily on the internal point distribution, which can select a bad principal axis.



Continuous covariance

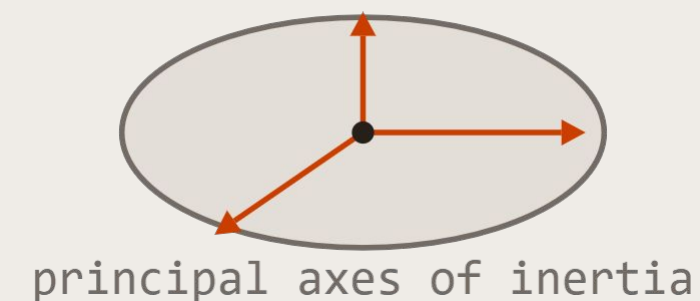
Integrate covariance across each primitive's whole face, not its vertices — the shape, not the tessellation. [Gottschalk00]



$O(n)$ with a hull $O(n \log n)$
without

Inertia tensor

Integrate over the volume of a uniform-density solid for the 3×3 inertia tensor; its eigenvectors are the principal axes. [Mirtich96a]



That last one is the same inertia tensor our dynamics module needs. Geometry and physics are computing the same object — compute it once.

LECTURE 14 • ITERATIVE OBB OPTIMIZATION

Project, Re-fit, Repeat While It Shrinks

Three ways to start

All-principal-components box — all three axes from PCA.

Max-principal-component box — longest axis from PCA; the other two from the minimum bounding rectangle of the projected vertices.

Min-principal-component box — same, starting from the shortest axis.

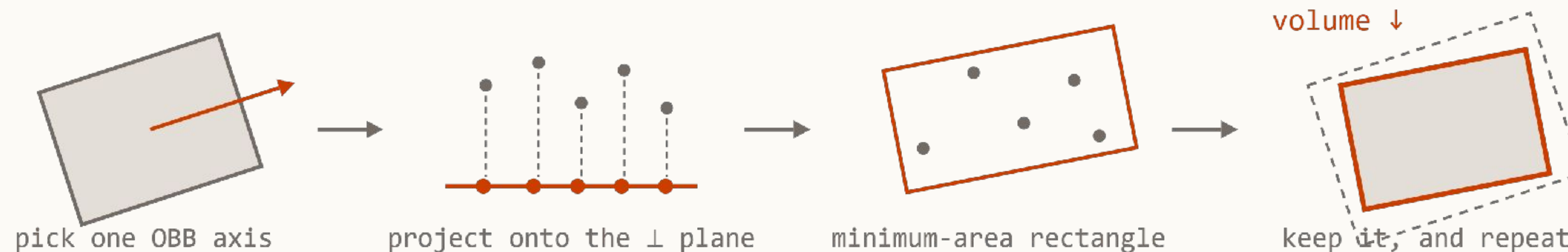
Then iterate

1 · Project all vertices onto the plane orthogonal to one axis of the OBB.

2 · Find the minimum bounding rectangle of the projected points.

3 · Take that rectangle's two axes as the new OBB axes.

4 · Repeat while the volume keeps shrinking.

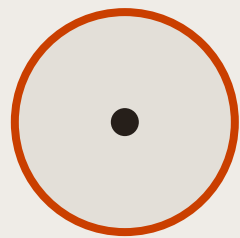


Second time today: a mediocre analytic start plus cheap iteration beats sophisticated analysis.

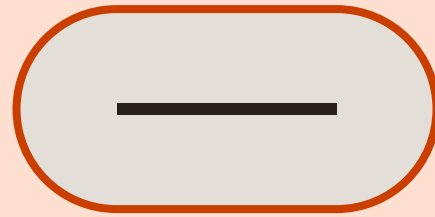
LECTURE 14 • SPHERE-SWEPT VOLUMES

A Volume Shaped Like the Tool You Are Holding

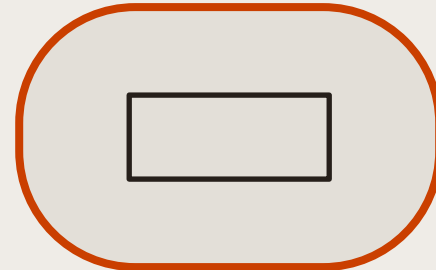
The Minkowski sum of an inner primitive and a sphere. All points within distance r of the inner medial structure belong to the volume.



point →
sphere



segment →
capsule



rectangle
→ lozenge

The cylinder, by contrast, is not a good bounding volume — flat caps and rims make its tests awkward.

Every SSV test has the same two steps

- 1 · Compute the distance between the inner structures.
- 2 · Compare it against the sum of the radii.

The only difference between any two sphere-swept tests is which distance calculation you use for the inner structures.

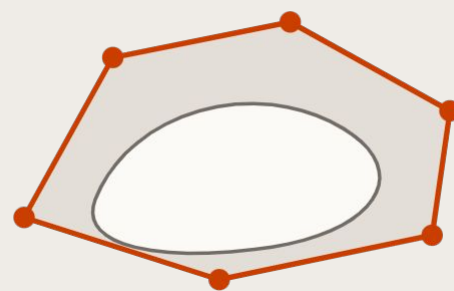
Look at the stylus in the user's hand: a shaft with a rounded tip. That is a capsule — segment-to-segment distance and one comparison, with no realignment and no fifteen axes. The haptic probe is the case SSVs were made for.

LECTURE 14 • HALFSPACE INTERSECTION VOLUMES

The Convex Hull Is Tightest — and Unaffordable

Why not just use the hull

Convex hulls form the tightest bounding volumes, but they are expensive to compute, take large amounts of memory, and are costly to operate upon.



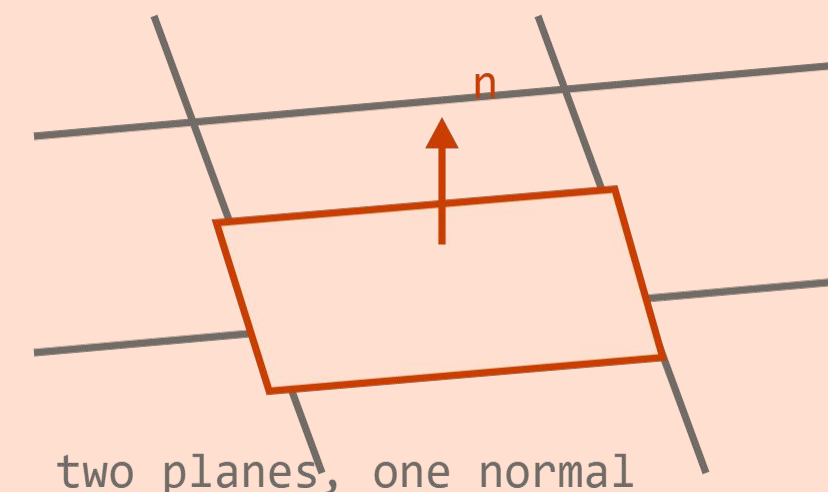
tight, and expensive in every other way

Limit the number of halfspaces and you get simpler alternatives that trade a little tightness for a lot of speed.

The slab

The infinite region of space between two parallel planes — a unit normal vector plus two signed distances from the origin.

Three slabs close a 3D volume. Add more and the fit approaches the convex hull arbitrarily closely.



two planes, one normal

three slabs, and the region they close

The catch: slab-based volumes give fast *ray* intersection tests but cannot easily be used as-is for object-object testing. Unless — every object shares the same set of normals.

LECTURE 14 • DISCRETE-ORIENTATION POLYTOPES

One Dial From a Box to a Convex Hull

Slab-based volumes where the same normals are **shared among all k-DOPs** , with components limited to $\{-1, 0, 1\}$ and left unnormalized — so projecting onto them is addition, not dot products. Also called fixed-direction hulls.

k	Directions	Shape
 6	$(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)$	Exactly an AABB — we have been using a 6-DOP all along
 8	$(\pm 1, \pm 1, \pm 1)$	The eight diagonal directions
 12	$(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)$	The twelve edge directions
 14	6-DOP + 8-DOP	An AABB with its eight corners cut off
 18	6-DOP + 12-DOP	An AABB with its twelve edges cut off · 20 and 26 continue the pattern

LECTURE 14 • K-DOPS IN PRACTICE

Excellent on Paper, Wrong for Our Loop

What they win

Overlap tests about an **order of magnitude faster than OBBs**, even for large k , thanks to the fixed direction planes.

An OBB's memory footprint corresponds roughly to a 14-DOP.

Probably tighter than OBBs, certainly as k grows and the polytope approaches the hull.

What they cost

Even when volumes rarely collide, the k -DOPs **must still be updated** as objects move — and that operation is expensive. The AABB's disease with k slabs instead of three.

For geometry poorly aligned with the fixed axes, OBBs can still be tighter — and OBBs perform better in close proximity, which is where a haptic contact lives.

Best when few dynamic objects are tested against many static ones, or when one object is used in many tests so the update amortizes.

Our tool is dynamic and rotating at 1 kHz — the amortization never happens. Hence the verdict in the notes: **not very popular for haptic rendering**.

LECTURES 13-14 • THE CATALOGUE, SCORED

Nobody Wins a Whole Row

Volume	Test cost	Fit	Memory	Rotation	Verdict for a 1 kHz loop
 Sphere	lowest	poor on long objects	4 floats	free — translate only	Top of the hierarchy; hard to beat as a first rejection
 AABB / 6-DOP	very low	fair	6 floats	must be realigned	Excellent for the static scene, awkward for the moving tool
 OBB	high — up to 15 axes	good	15 floats	free — rotate the frame	Best in close proximity; costliest exactly when contact is deep
 Capsule / lozenge	low — one distance	good on shafts	small	free	The shape of the probe itself — our default for the tool
 k-DOP	low for its tightness	tightest short of a hull	≈ OBB at k=14	expensive update, always	Few dynamic vs. many static objects — not our case
 Convex hull	highest	tightest	largest	free	The bound the others approximate, not a runtime option

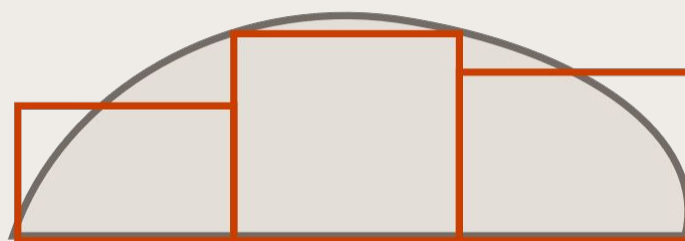
WHAT THIS DECIDES IN OUR BUILD

Different Volumes for the Hand and the World

01

The scene is static — buy tightness

Computed once, offline, never realigned. Tight AABBs or OBBs, and cost of construction stops mattering.

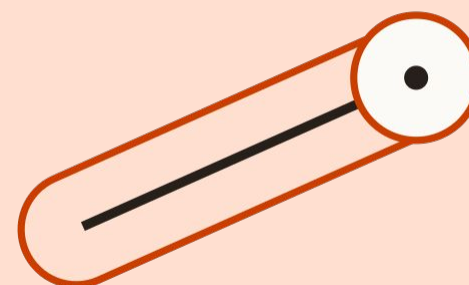


built once, offline

02

The tool moves — buy rotation invariance

A sphere at the tip, a capsule along the shaft. Update cost is charged 1,000 times a second, so make it zero.

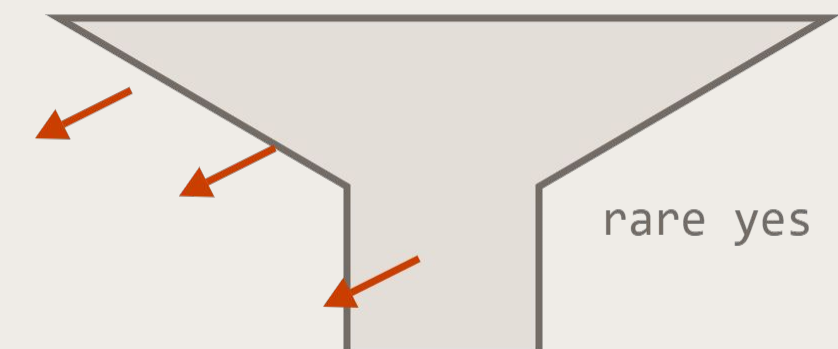


capsule shaft, sphere tip

03

Optimize the answer you get

Almost every cycle is a no. Design for the cost of rejection, and let the rare yes be slow.



no

rare yes

One multiplication survives: one capsule against 2×10^5 triangles is still 2×10^5 tests. **Next session** removes it — bounding volume hierarchies, and the traversal that turns a linear scan into a logarithmic one.

LECTURES 13–14 • RECAP

Four Things to Leave With

The budget

One millisecond, non-negotiable — a human closes this loop with their fingertips.

1 kHz

The cheap no

A bounding volume only ever proves absence. Build the pipeline around the answer you actually get.

reject, then descend

The five-way trade

Test cost, fit, construction, transform, memory. Haptics weights the transform column far more than graphics does.

Shape follows tool

Let the geometry you are wrapping pick the wrapper. For a probe with a rounded tip, that is a capsule.

Homework • L13: an example illustrating how `We1z1Sphere()` works, explaining its recursion step by step. Due in one week.

Reading • L14: Ericson, *Real-Time Collision Detection*, Ch. 4 — write a brief summary. Due in one week.